MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2026

FOURTH SEMESTER
Discipline Specific Course (DSC) - MG4DSEBTG201 - Biostatistics

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A - One word answer - Answer any 10 questions. Each question carries 2 marks

1. Differentiate primary data and secondary data. [U]/CO2
2. Define frequency distribution? [K]/CO2
3. Define histogram? Identify how is it drawn and when? [U]/CO3
4. Discuss the demerits of arithmetic mean? [U]/CO3
5. Explain the important characteristics that a good average should possess. [U]/CO3
6. Why standard deviation considered being the most popular measure of dispersion [U]/CO3
7. Recall the essential qualities of a good sample? [K]/CO1
8. Give two examples of poisson distribution. [K]/CO3
9. Explain Chi-square test. [U]/CO3
10. Define critical value? [U]/CO3
11. Define is bubble sort? [U]/CO4
12. Name any two statistical packages used for data analysis. [U]/CO4

(10 x 2 = 20)

Part B- Short Essay- Answer any 4 questions. Each question carries 5 marks

13. Define statistics? Mention the characteristics of statistics? [K]/CO1
14. Define primary data. State the various methods of collecting primary data [U]/CO1
15. Discuss the characteristics of classification? [U]/CO2
16. Describe the different components of frequency distribution? [U]/CO2
17. Describe MS Excel as a statistical software. Explain its main components and features. [U]/CO4
18. Discuss the different probability sampling techniques used in research. [U]/CO3

(4 x 5= 20)

Part C- Essay- Answer any 1 question. Each question carries 10 marks

19. Give an account of measurers of central tendency. [K]/CO2
20. Discuss briefly the graphical methods of representing a frequency distribution. [U]/CO2
21. State probability theory? Point out difference between binomial, poisson and normal distribution. [U]/CO3
22. Apply an appropriate statistical test to compare the means of two samples, explain the procedure, and interpret the result. [A]/CO3

(2X15=30)